Usmonov Mukhammad

AI Engineer | Python Backend

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SUMMARY

AI Engineer with strong Python backend development experience and hands-on expertise in LLM applications, Retrieval-Augmented Generation (RAG), vector databases, and AI agents. Built production-ready AI systems using FastAPI, LangChain, Qdrant, Hugging Face, and leading AI APIs. Experienced in machine learning fundamentals, backend architecture, and deploying scalable AI-powered applications. Passionate about applying AI technologies to create practical solutions and advance intelligent systems.

EXPERIENCE

Python Developer — Jul '25 — Jan '26

Ministry of Economy and Finance of Republic of Uzbekistan — Tashkent

- Developed and deployed a URL shortening microservice using MD5 hashing to generate shortened URLs, enabling efficient link management for SMS platforms and improving campaign tracking and distribution.
- Designed RESTful APIs for datasets exceeding 100k records, implementing pagination and query optimization to reduce response latency by 40%.
- Built a Kafka-based centralized logging pipeline handling 50K+ events daily, enabling real-time monitoring and faster incident resolution.
- Participated in backend architecture design and code review processes, helping improve code quality and maintainability across multiple services.

PROJECTS

PostgresPro Support Link — May '26 — Jun '26, Tashkent

- Engineered a production-ready RAG system for PostgreSQL 14 documentation using LangChain, Qdrant, Hugging Face embeddings, and Groq LLMs.
- Developed a Streamlit-based AI assistant capable of answering PostgreSQL-related questions with context-aware and source-grounded responses.
- Built a semantic retrieval pipeline, implementing document preprocessing, chunking, embedding generation, and vector similarity search.
- Integrated large language models through the Groq API to improve answer quality and reduce hallucinations using retrieval-augmented generation.
- Leveraged PostgreSQL documentation as a domain-specific knowledge base, enabling accurate technical support and knowledge discovery.

AI Medicine Reminder System Link — Apr '26 — May '26, Tashkent

- Developed an AI-powered medicine reminder system using FastAPI, Aiogram, and APScheduler to automate medication scheduling and user notifications.
- Built a Telegram-based interface that enables users to manage medication plans and receive timely reminder alerts.
- Integrated Gemini API to extract medicine names from prescription images, automating data entry through multimodal AI processing.
- Leveraged Cerebras AI API to validate, normalize, and enhance extracted medication information, improving recognition accuracy and user experience.
- Designed backend services and scheduling workflows to ensure reliable reminder delivery and efficient medication management.

EDUCATION

B.Sc. in Computer Science(AI Solutions and Applications), PDP University — Tashkent

TECHNICAL SKILLS

Languages Python, C++
AI / LLM LLM, RAG, OpenAI API, Anthropic(Claude) API, Google Gemini API, Groq API, CerebrasAI API, EasyOCR API, Ollama, Tokenization, n8n, TensorFlow, Keras, Pandas, ML
Backend FastAPI, Django, Django REST Framework, Ngnix, Docker
Databases Qdrant, PostgreSQL, MySQL, SQLite, MongoDB, Redis
Cloud GCP, AWS, Digital Ocean
Infrastructure & Tools Git, Linux, Postman, Swagger, SQLAlchemy ORM, Kafka, RabbitMQ, Celery, Jira

Agentic Frameworks & Protocols MCP, Claude Code, Codex, Github Copilot, Antigravity, Cursor
Concepts REST APIs, System Design, Async Processing, Caching, Message Queues, OOP, gRPC, GraphQL, Microservices, JWT, Agile, Canban, Supervised Learning, Unsupervised Learning

COURSES

Machine Learning Specialization by Andrew Ng on Stanford online